

CARLOS MARTINEZ RUIZ

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PROFESSIONAL SUMMARY

Seasoned Machine Learning Engineer with 14 years of experience designing and deploying scalable ML systems in production environments. Specialized in deep learning, computer vision, and large-scale data processing with a proven track record of delivering high-impact solutions for Fortune 500 companies and innovative startups. Passionate about leveraging cutting-edge technologies to solve complex business problems while mentoring junior engineers and driving technical excellence across organizations.

WORK EXPERIENCE

Staff Machine Learning Engineer

Nexus AI Solutions

nexusai.io

Barcelona, Spain

March 2021 - Present

Led the development and deployment of a computer vision pipeline processing 500M+ images daily, improving object detection accuracy by 23% and reducing inference latency by 40%.

Architected a distributed training framework supporting models with over 100B parameters, enabling the team to train 5x faster and reduce infrastructure costs by 35%.

Mentored 8 junior and senior machine learning engineers, establishing best practices for model development, experimentation tracking, and code review processes.

Designed and implemented an automated ML monitoring system detecting model drift and performance degradation in real-time, preventing 12+ production incidents annually.

Senior Machine Learning Engineer

Quantum Data Systems

www.quantumdatasystems.net

Barcelona, Spain

July 2017 - February 2021

Built end-to-end recommendation systems serving personalized content to 15M+ monthly active users with A/B testing infrastructure reducing churn by 8%.

Developed NLP models for multilingual text classification and sentiment analysis achieving 94% accuracy across 12 languages using transformer-based architectures.

Established MLOps infrastructure including model versioning, containerization, and CI/CD pipelines that reduced deployment time from weeks to hours.

Collaborated with product and data teams to define KPIs and success metrics, translating business requirements into machine learning objectives that drove 2.5M EUR in incremental revenue.

Machine Learning Engineer

Google Spain

www.google.es

Madrid, Spain

January 2010 - June 2017

Developed machine learning models for Google Search ranking and relevance, impacting search quality for hundreds of millions of daily queries across multiple languages.

Implemented feature engineering pipelines and conducted extensive experimentation using statistical methods, processing terabytes of structured and unstructured data.

Optimized model inference serving latency and memory consumption, enabling deployment across Google's global infrastructure serving 99.99% uptime requirements.

Contributed to internal machine learning frameworks and libraries used by hundreds of engineers across the company, improving productivity and code quality standards.

EDUCATION

Master of Science in Computer Science

Universidad Politecnica de Cataluna (UPC)

Barcelona, Spain

2008

Bachelor of Science in Computer Science

Universidad Autonoma de Madrid (UAM)

Madrid, Spain

2006

TECHNICAL SKILLS

Machine Learning: Deep Learning, Computer Vision, NLP, Recommendation Systems, Time Series Analysis, Reinforcement Learning, Transfer Learning

Programming Languages: Python, Scala, Java, C++, SQL

Frameworks and Libraries: TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, JAX

Cloud Platforms: Google Cloud Platform, AWS, Azure

Data Processing: Apache Spark, Hadoop, Apache Flink, Airflow, Kafka

MLOps: Model Serving, Docker, Kubernetes, Model Monitoring, Experiment Tracking, DVC

Databases: PostgreSQL, MongoDB, BigQuery, Elasticsearch, Redis

Other: Git, Linux, Jupyter, Pandas, NumPy, Matplotlib, Statistical Analysis

LANGUAGES

Spanish (Native)

English (Fluent)

Catalan (Fluent)